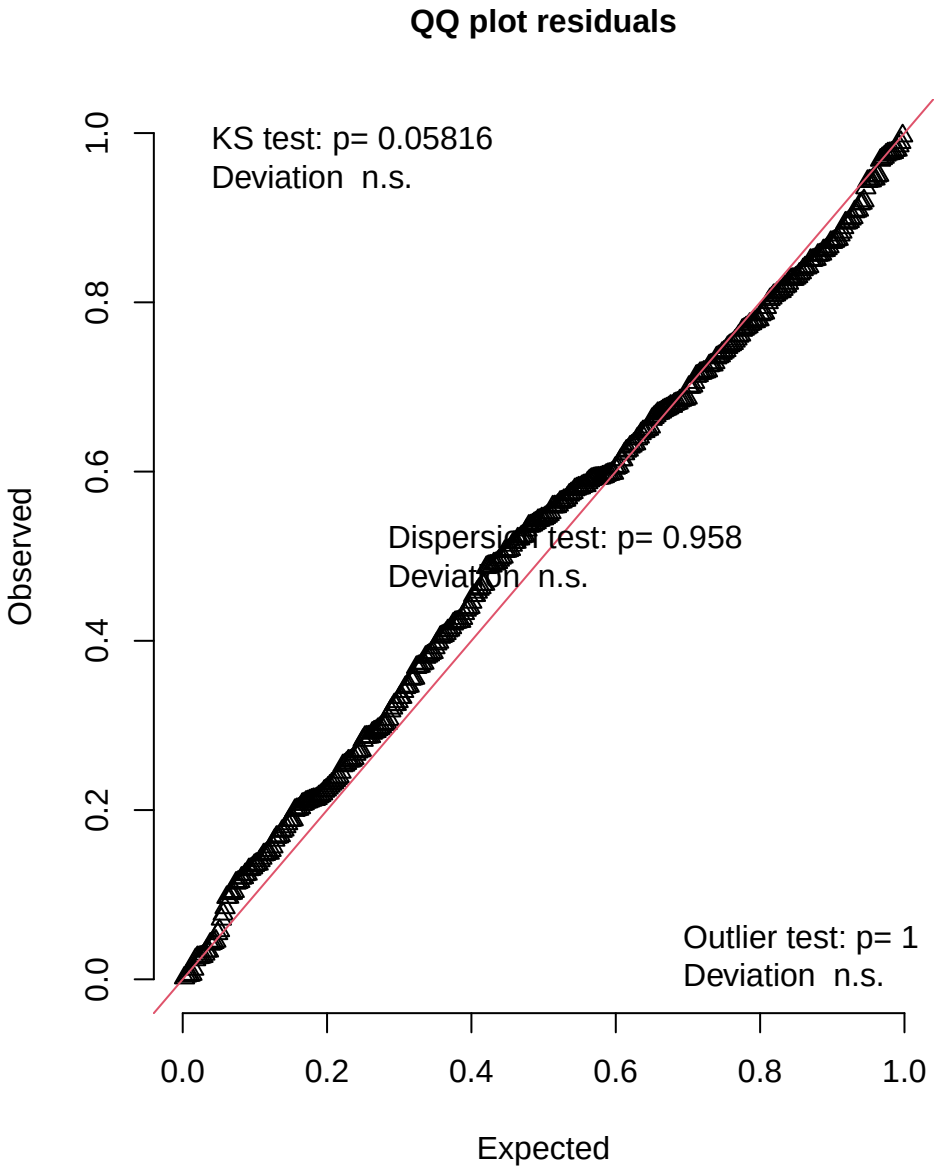
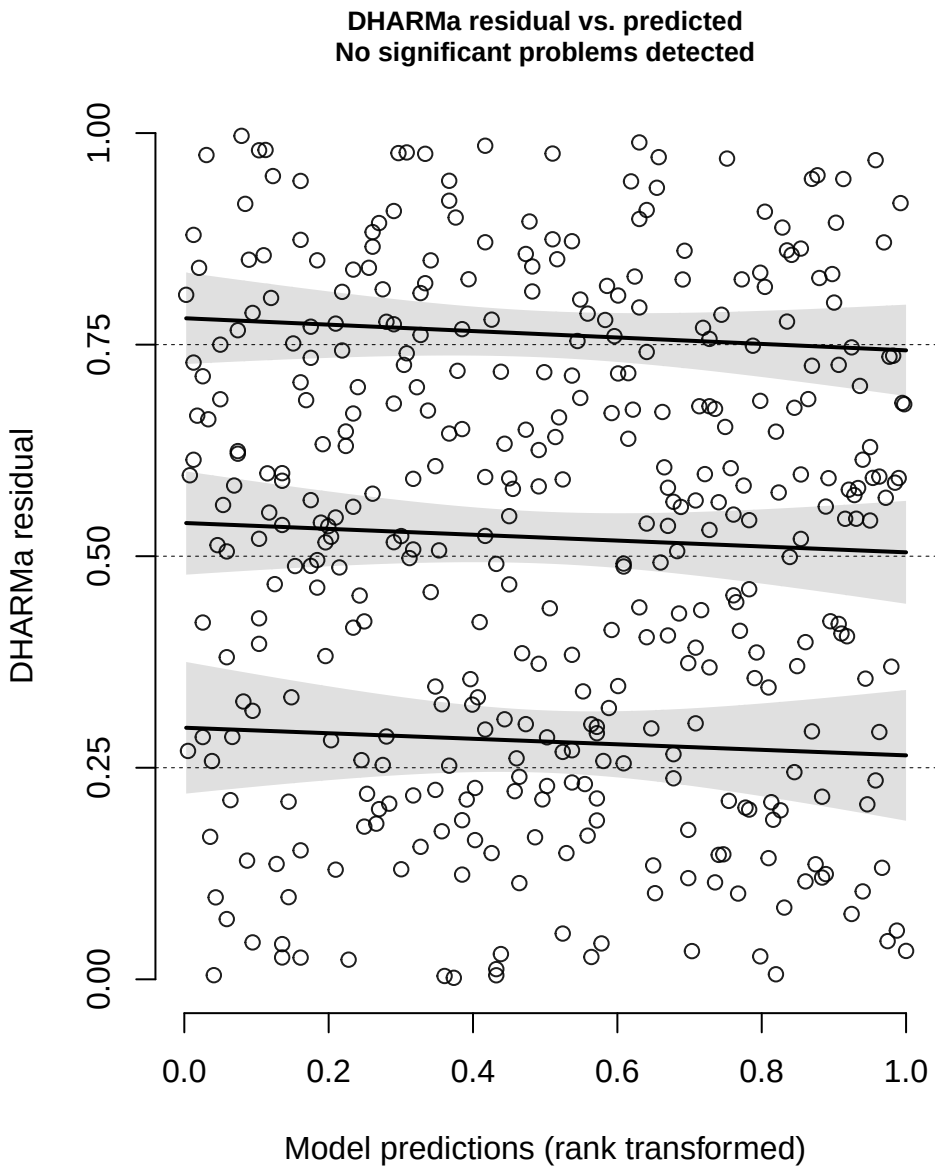
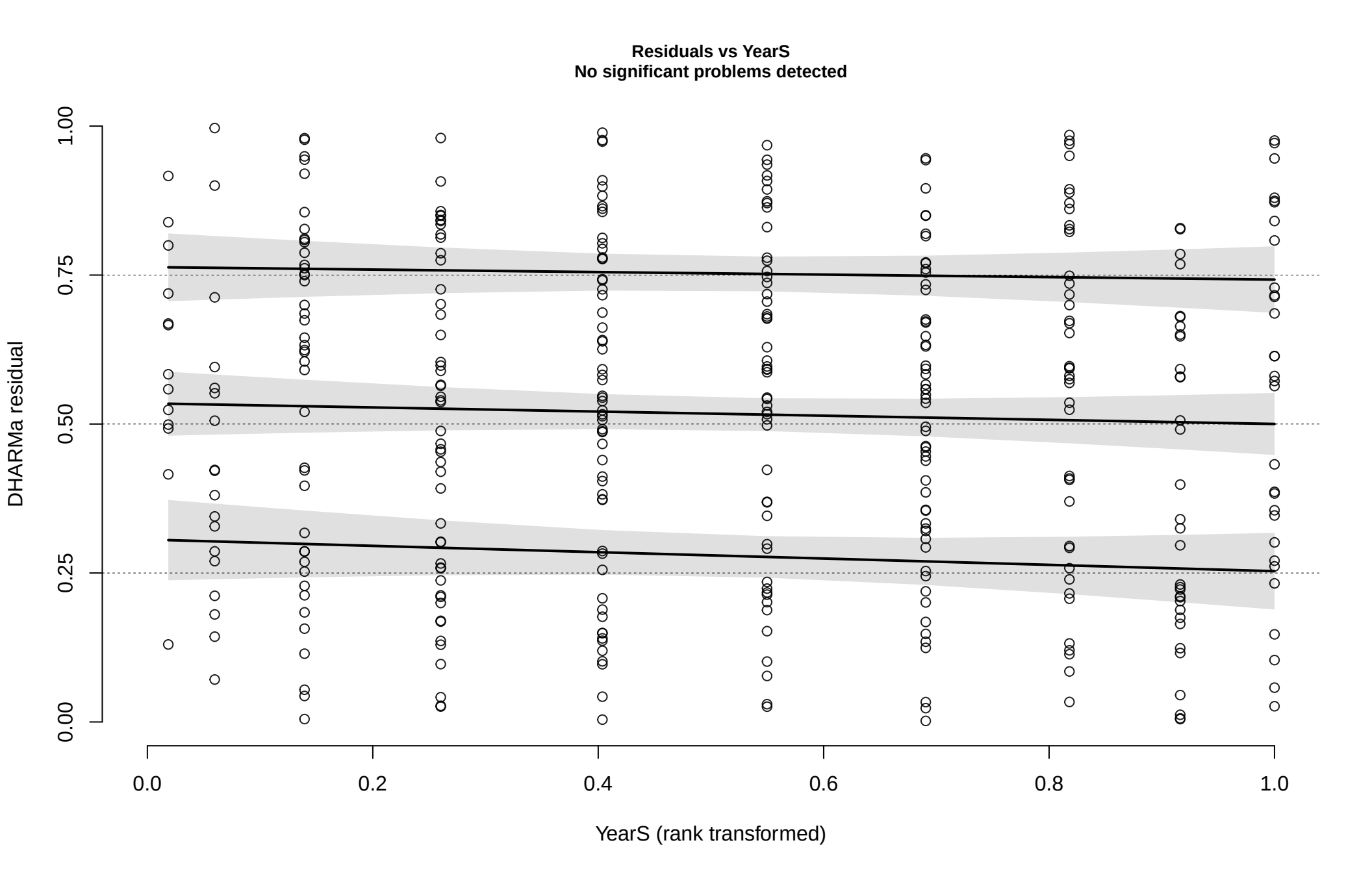
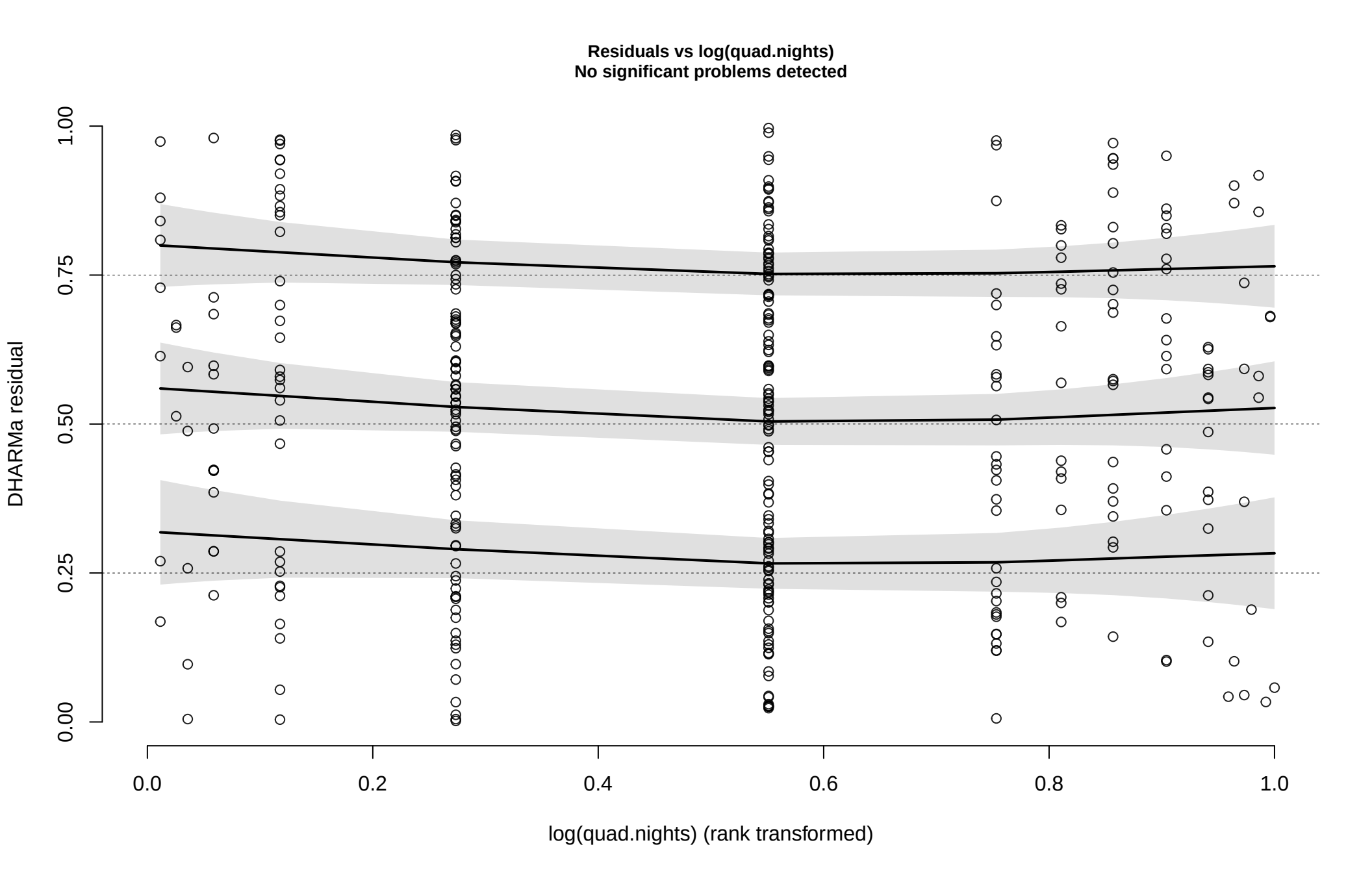
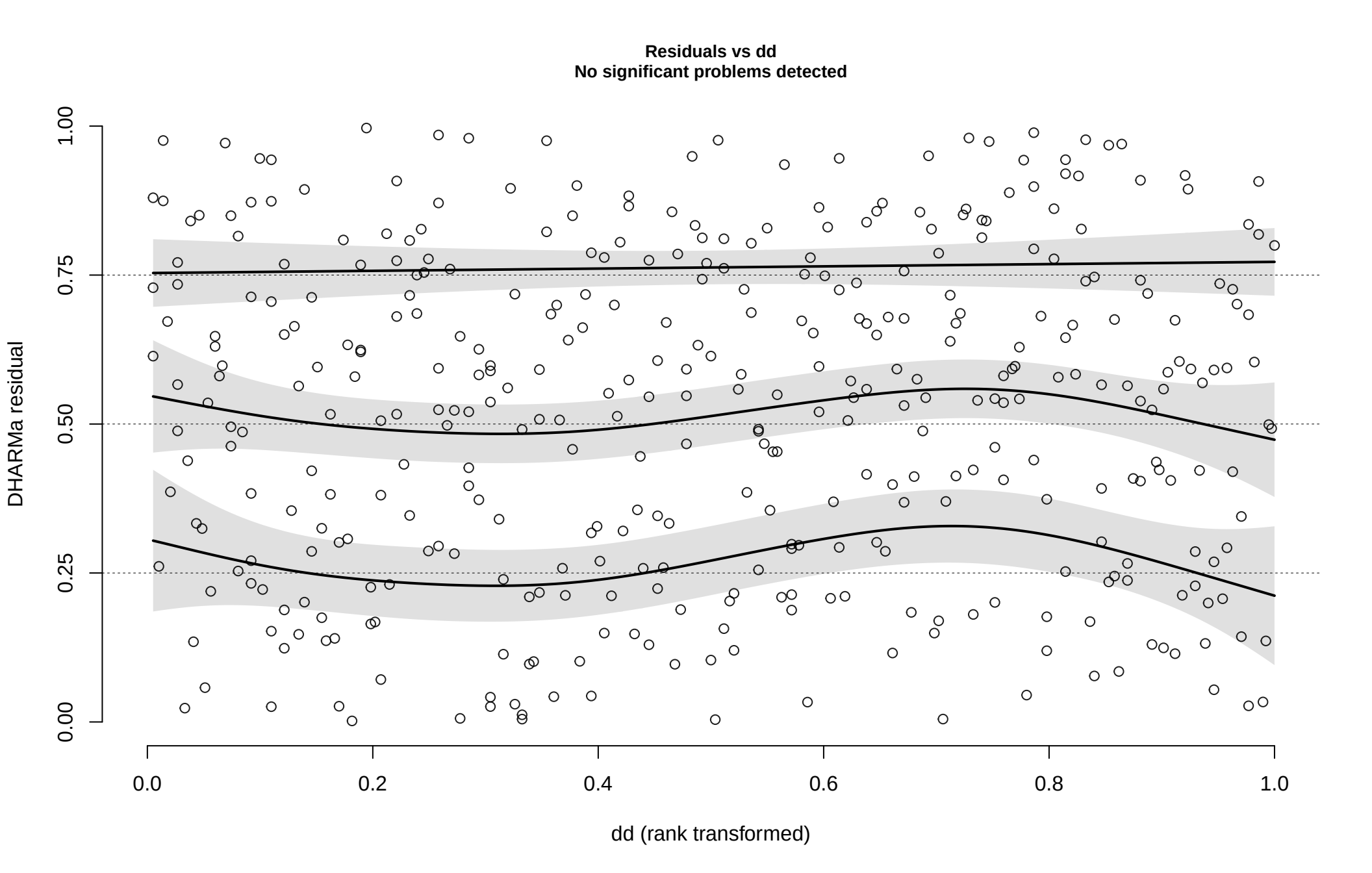
DHARMa residuals: South Coastal Region_COTO

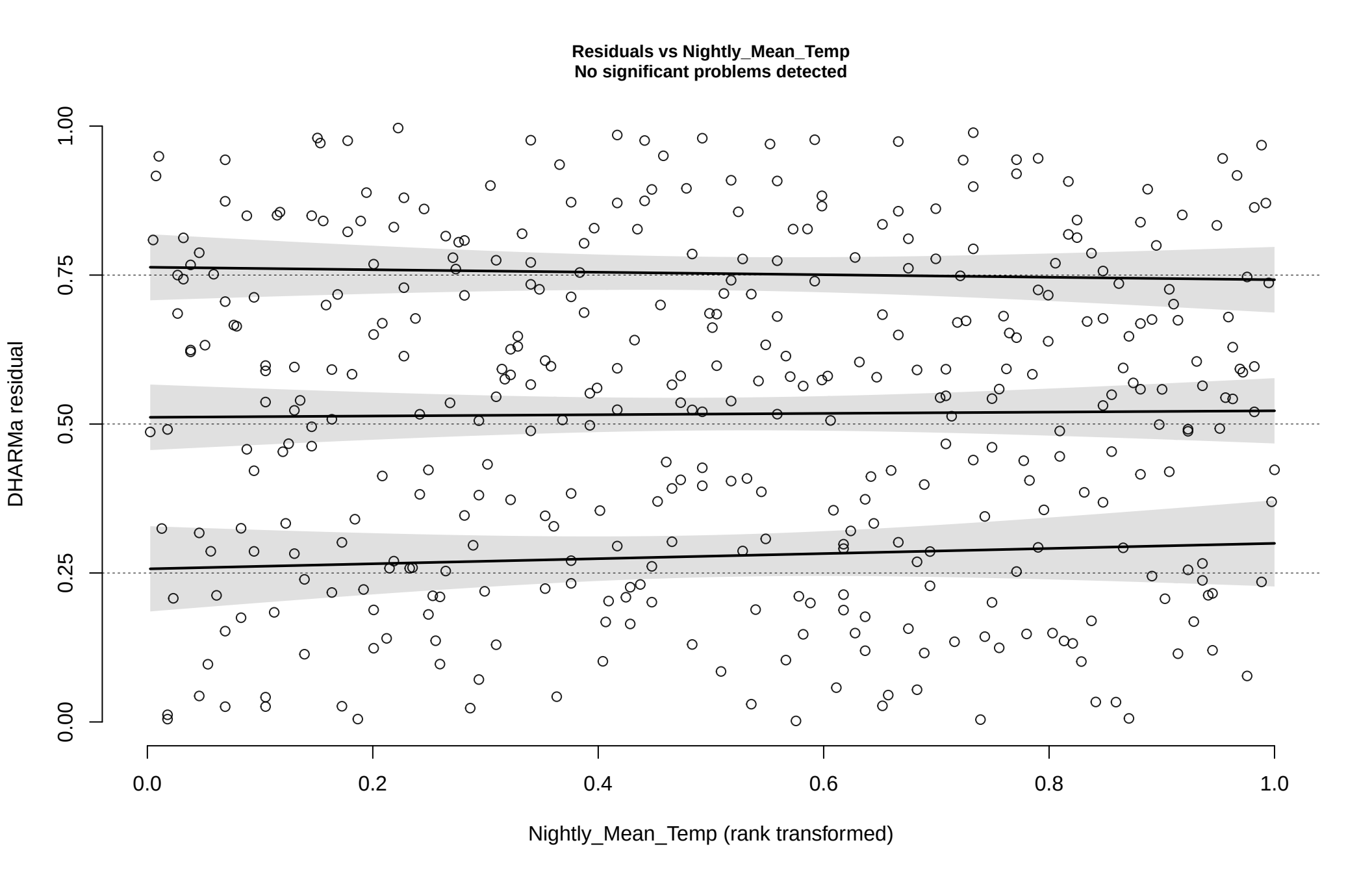


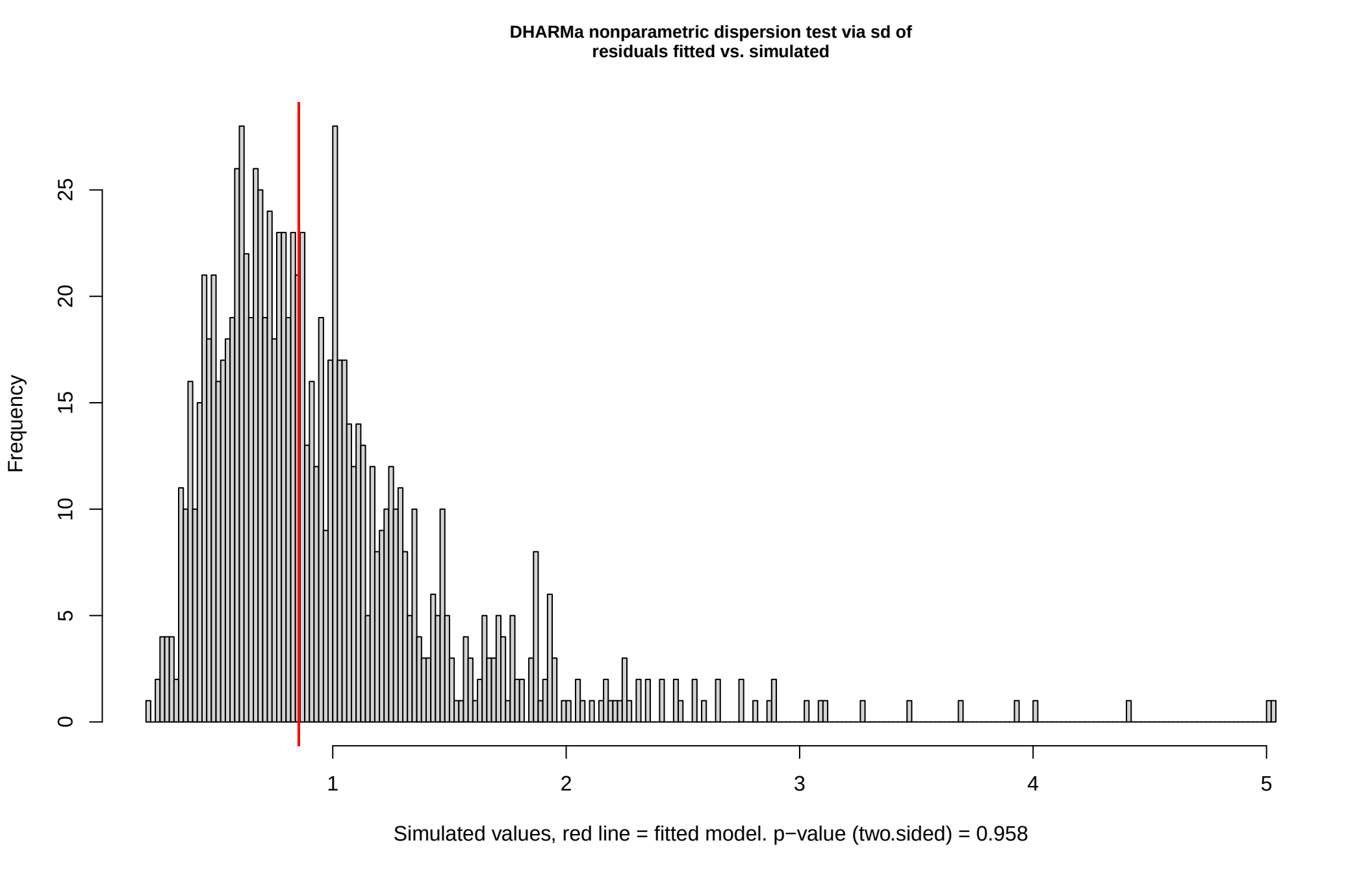


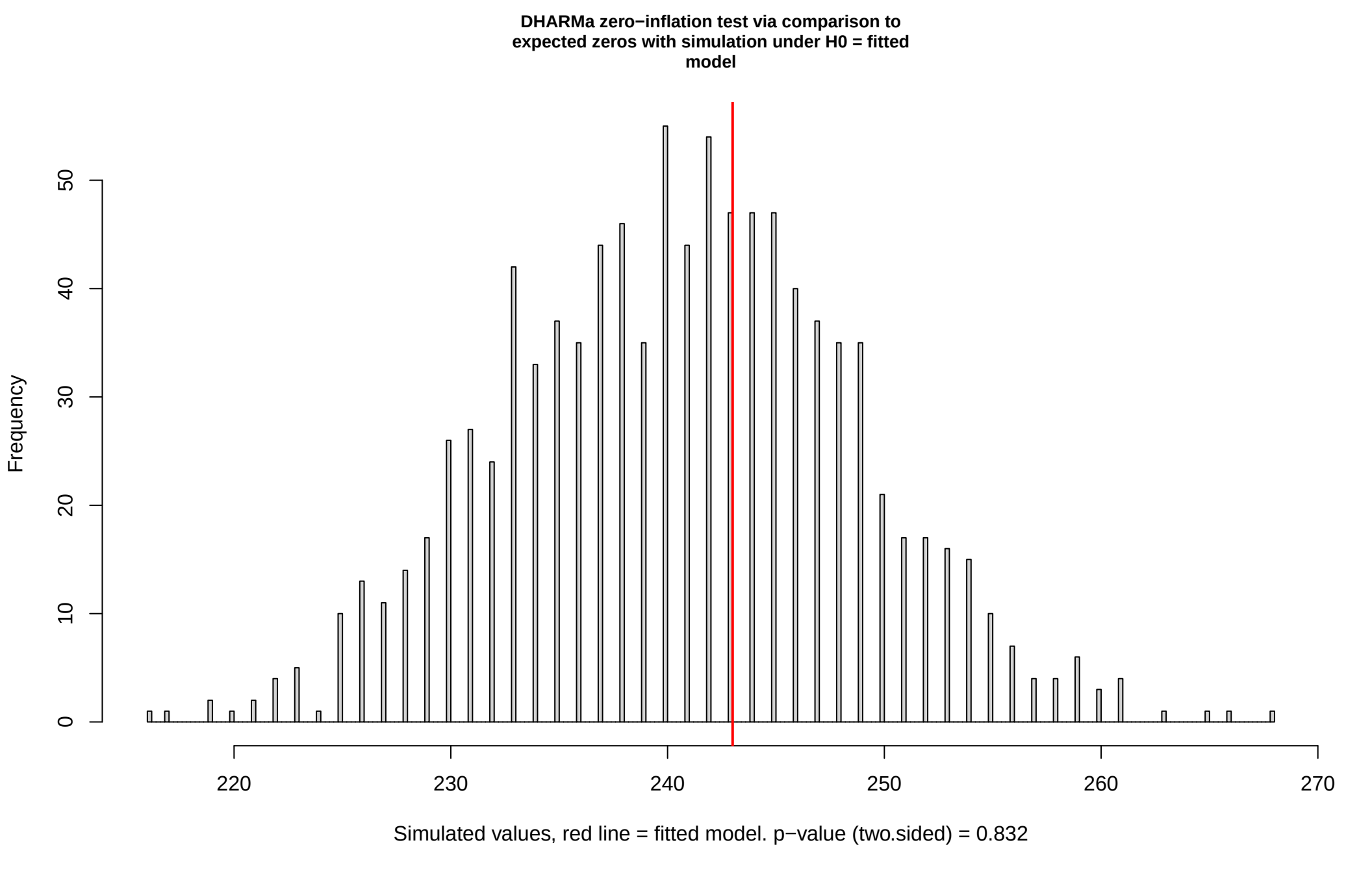

Residuals vs YearS
No significant problems detected
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
YearS (rank transformed)

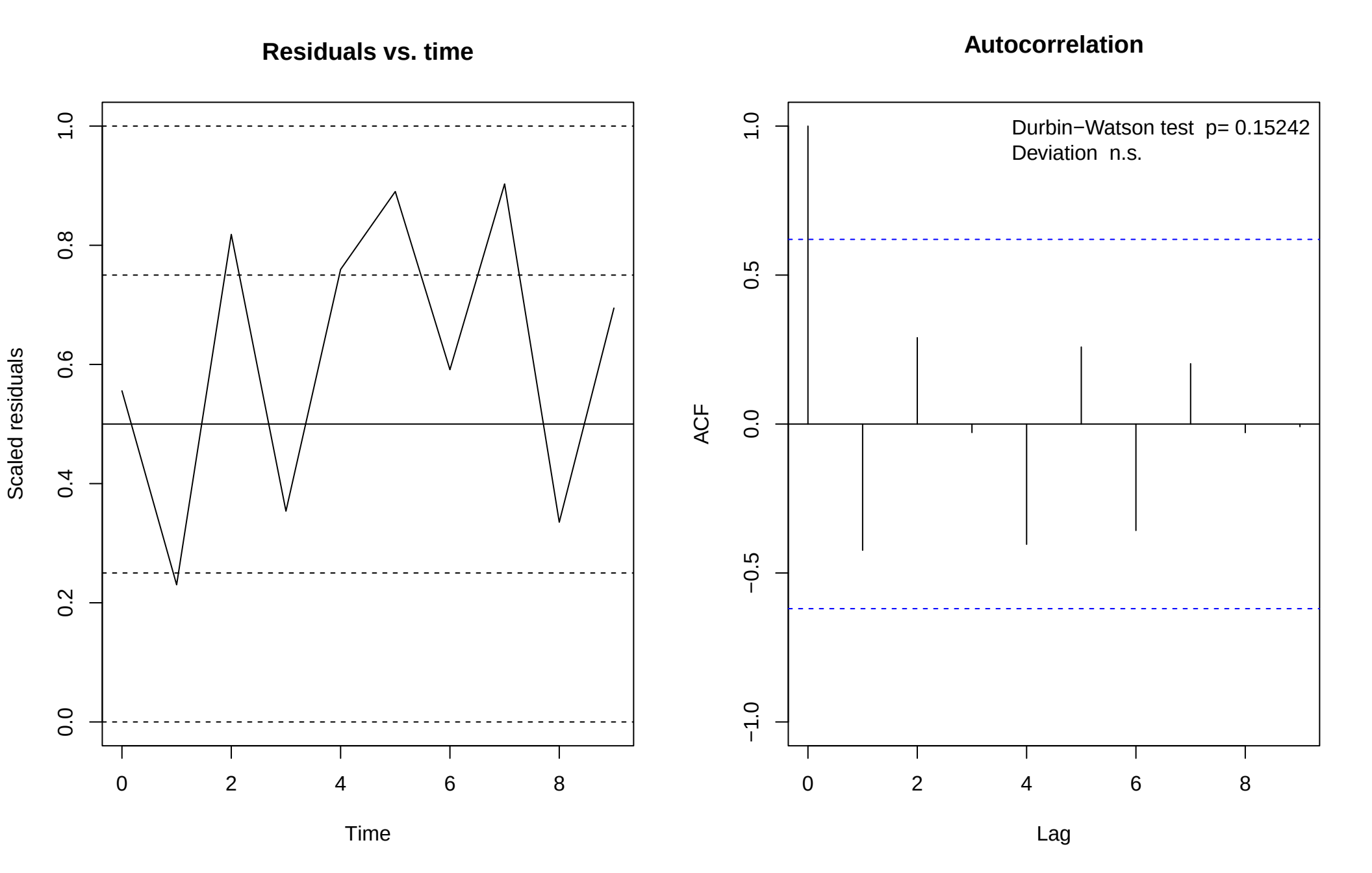

Residuals vs log(quad.nights)
No significant problems detected
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
log(quad.nights) (rank transformed)

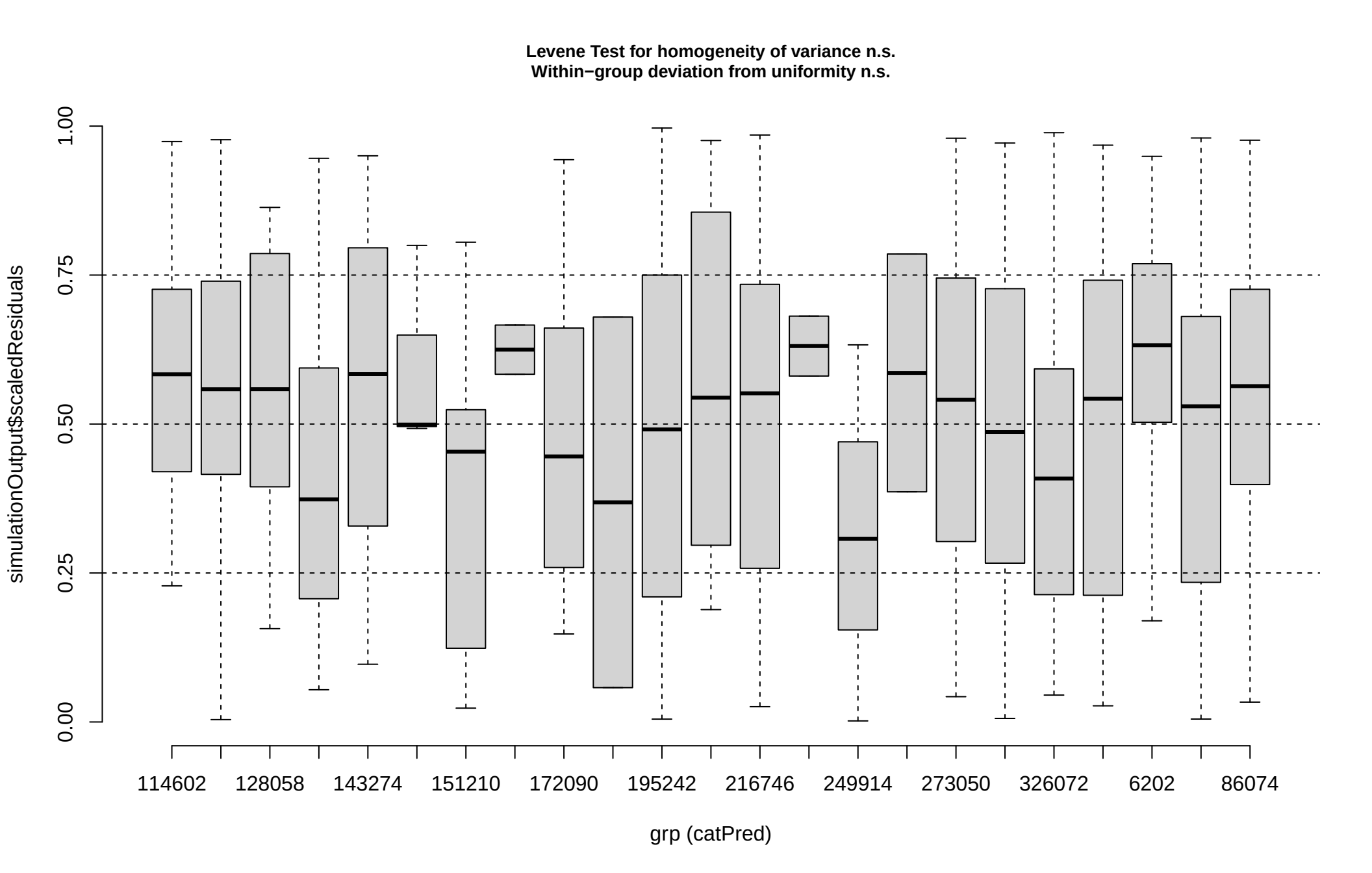

Residuals vs dd
No significant problems detected
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
dd (rank transformed)

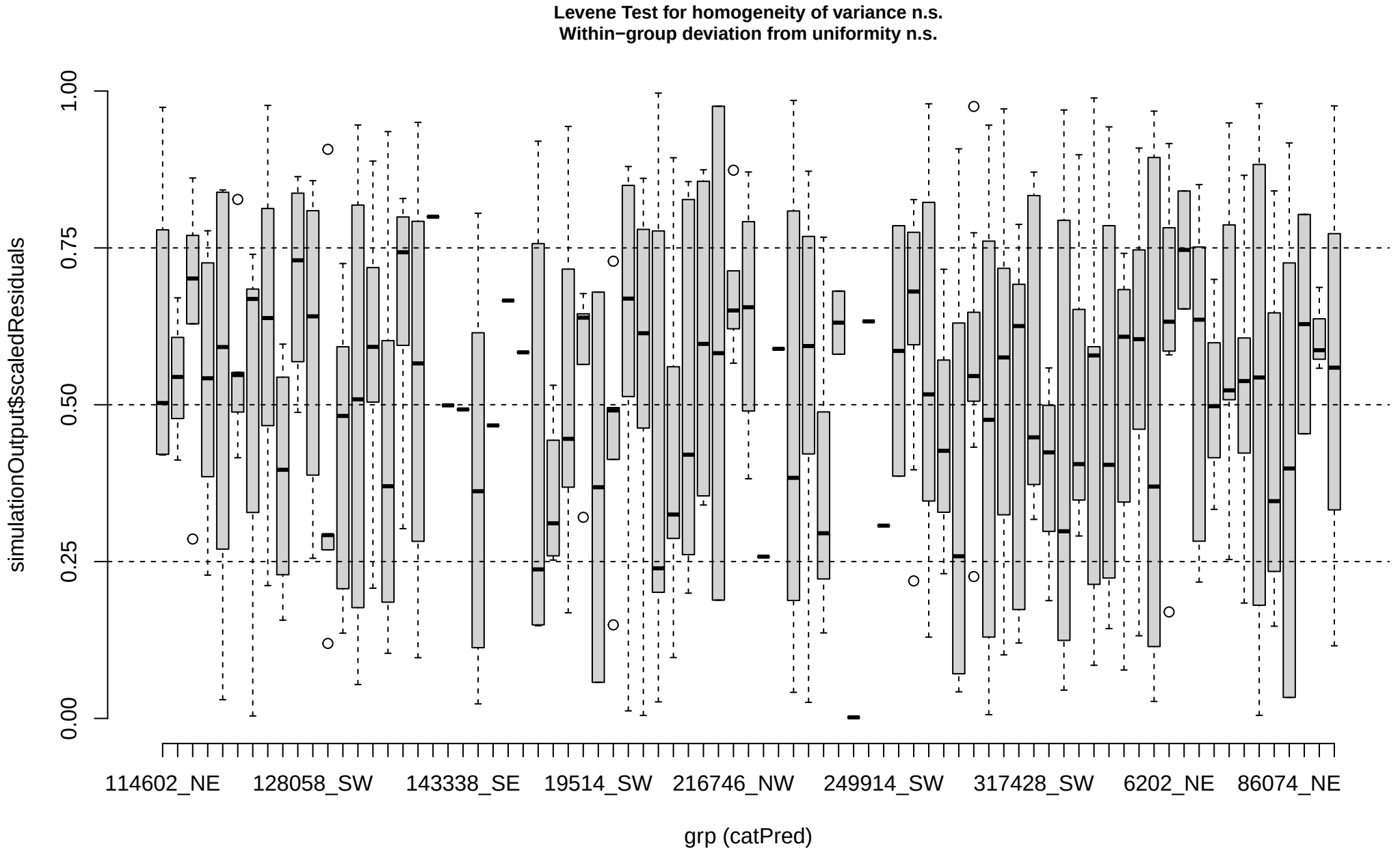

Residuals vs Nightly_Mean_Temp
No significant problems detected
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
Nightly_Mean_Temp (rank transformed)

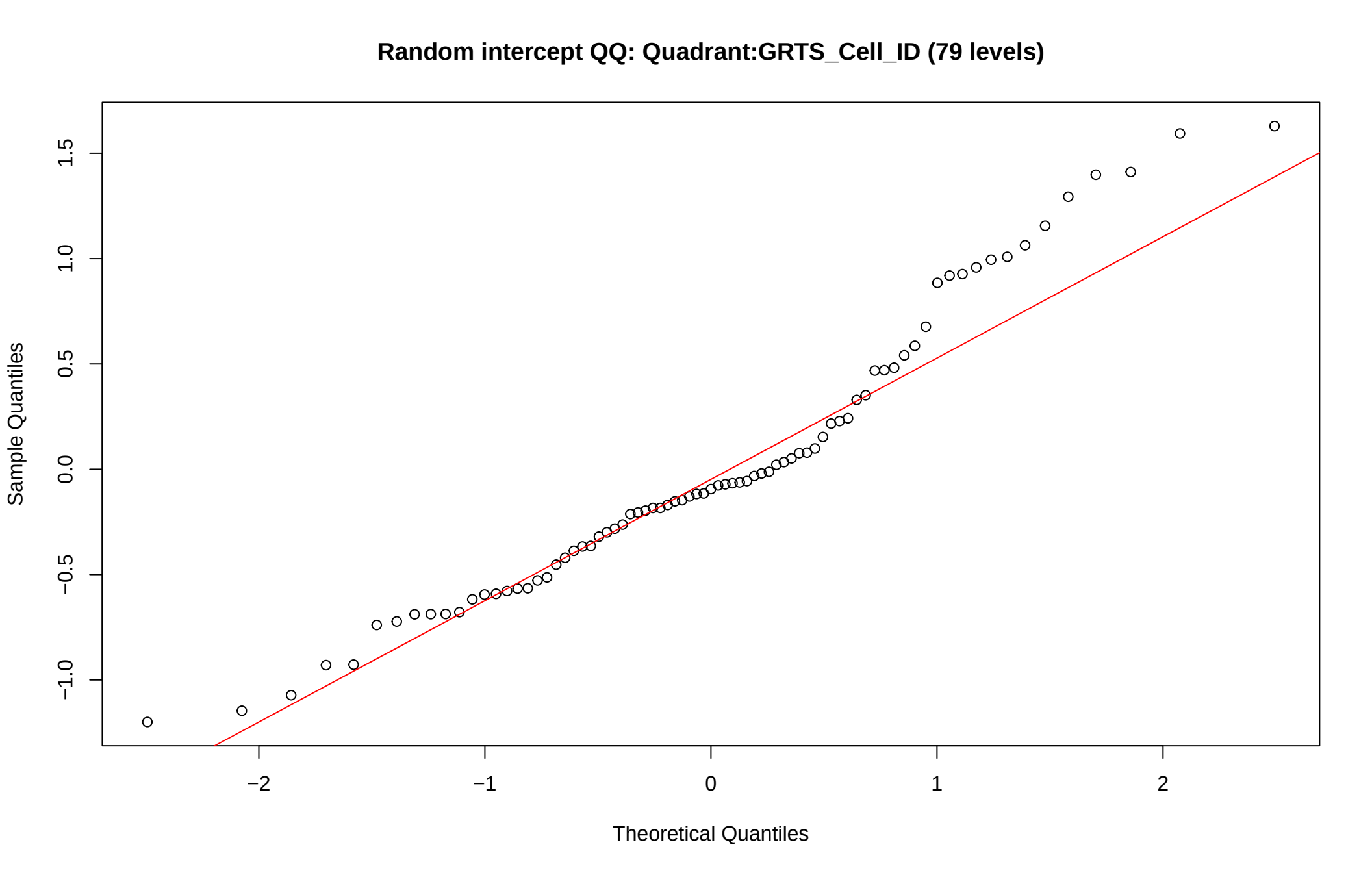

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated
Frequency
0
5
10
15
20
25
1
2
3
4
5
Simulated values, red line = fitted model. p−value (two.sided) = 0.958

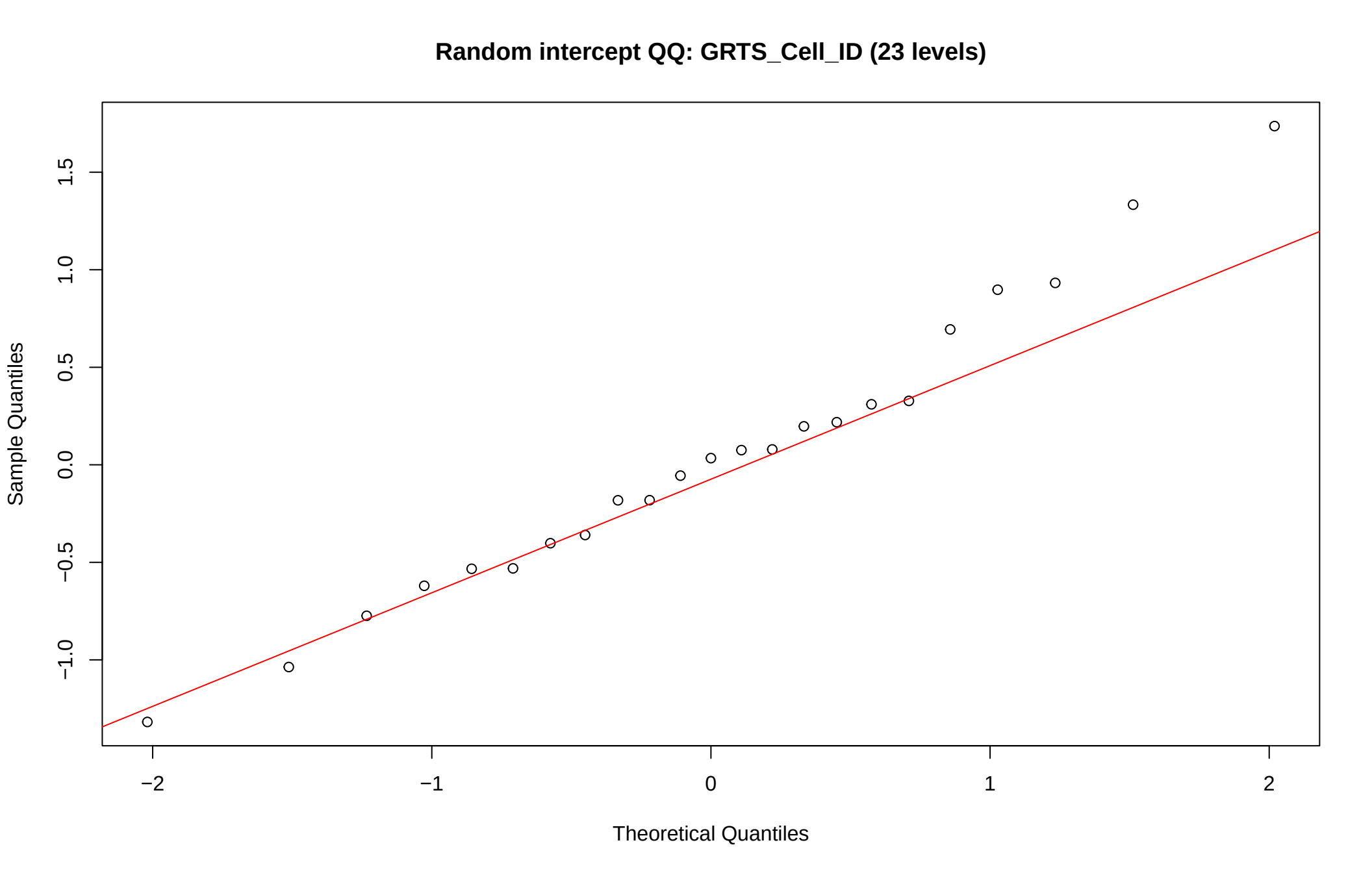

DHARMa zero−inflation test via comparison to expected zeros with simulation under H0 = fitted model
Frequency
0
10
20
30
40
50
220
230
240
250
260
270
Simulated values, red line = fitted model. p−value (two.sided) = 0.832


Residuals vs. time
Scaled residuals
0.0
0.2
0.4
0.6
0.8
1.0
0
2
4
6
8
Time
Autocorrelation
ACF
−1.0
−0.5
0.0
0.5
1.0
Durbin−Watson test p= 0.15242
Deviation n.s.
0
2
4
6
8
Lag


Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
0.00
0.25
0.50
0.75
1.00
114602
128058
143274
151210
172090
195242
216746
249914
273050
326072
6202
86074
grp (catPred)


Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
1.00
0.75
0.50
0.25
0.00
114602_NE
128058_SW
143338_SE
19514_SW
216746_NW
249914_SW
317428_SW
6202_NE
86074_NE
grp (catPred)


Random intercept QQ: Quadrant:GRTS_Cell_ID (79 levels)
Sample Quantiles
Theoretical Quantiles
−1.0
−0.5
0.0
0.5
1.0
1.5
−2
−1
0
1
2


Random intercept QQ: GRTS_Cell_ID (23 levels)
Sample Quantiles
Theoretical Quantiles
−1.0
−0.5
0.0
0.5
1.0
1.5
−2
−1
0
1
2